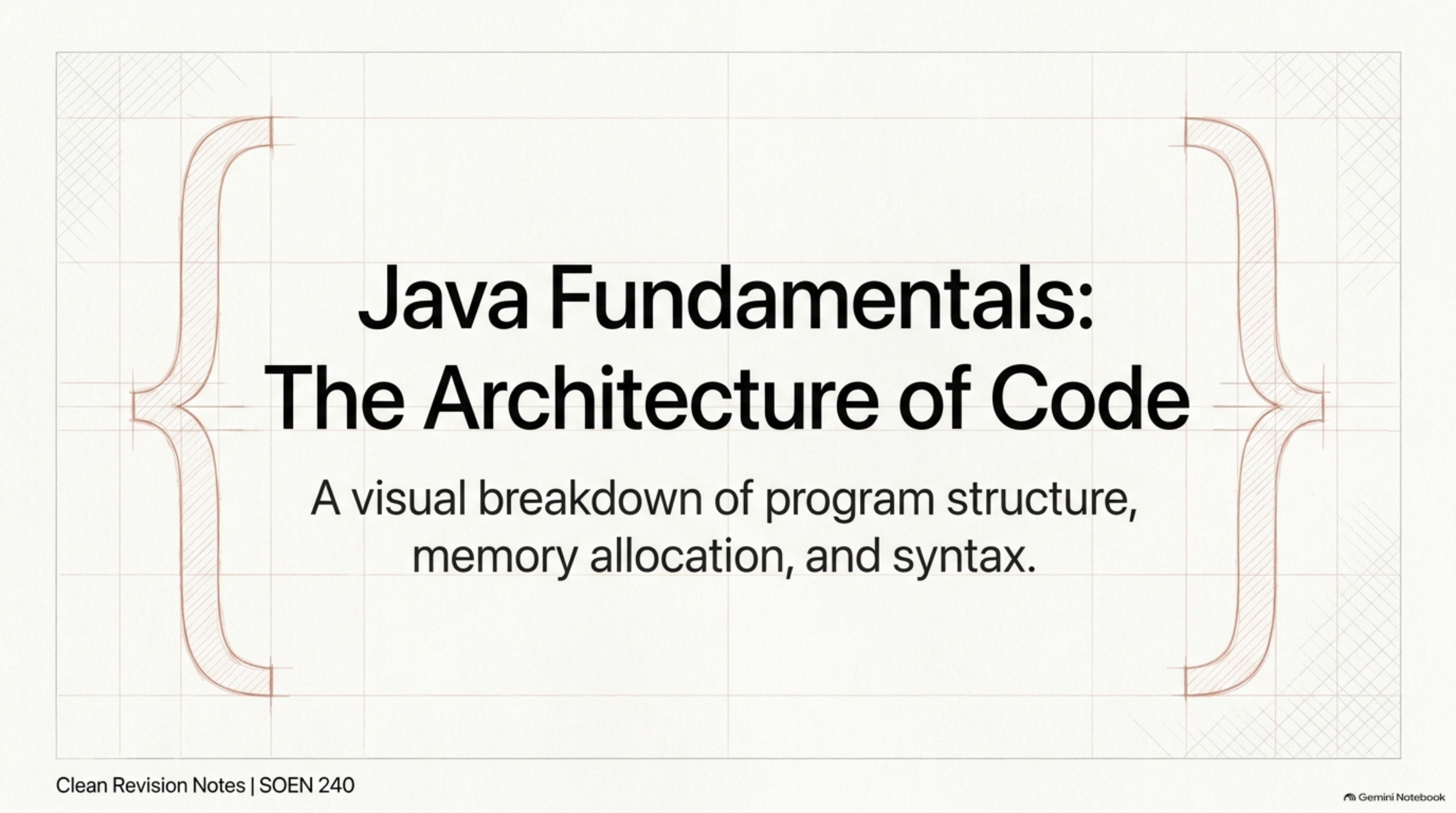


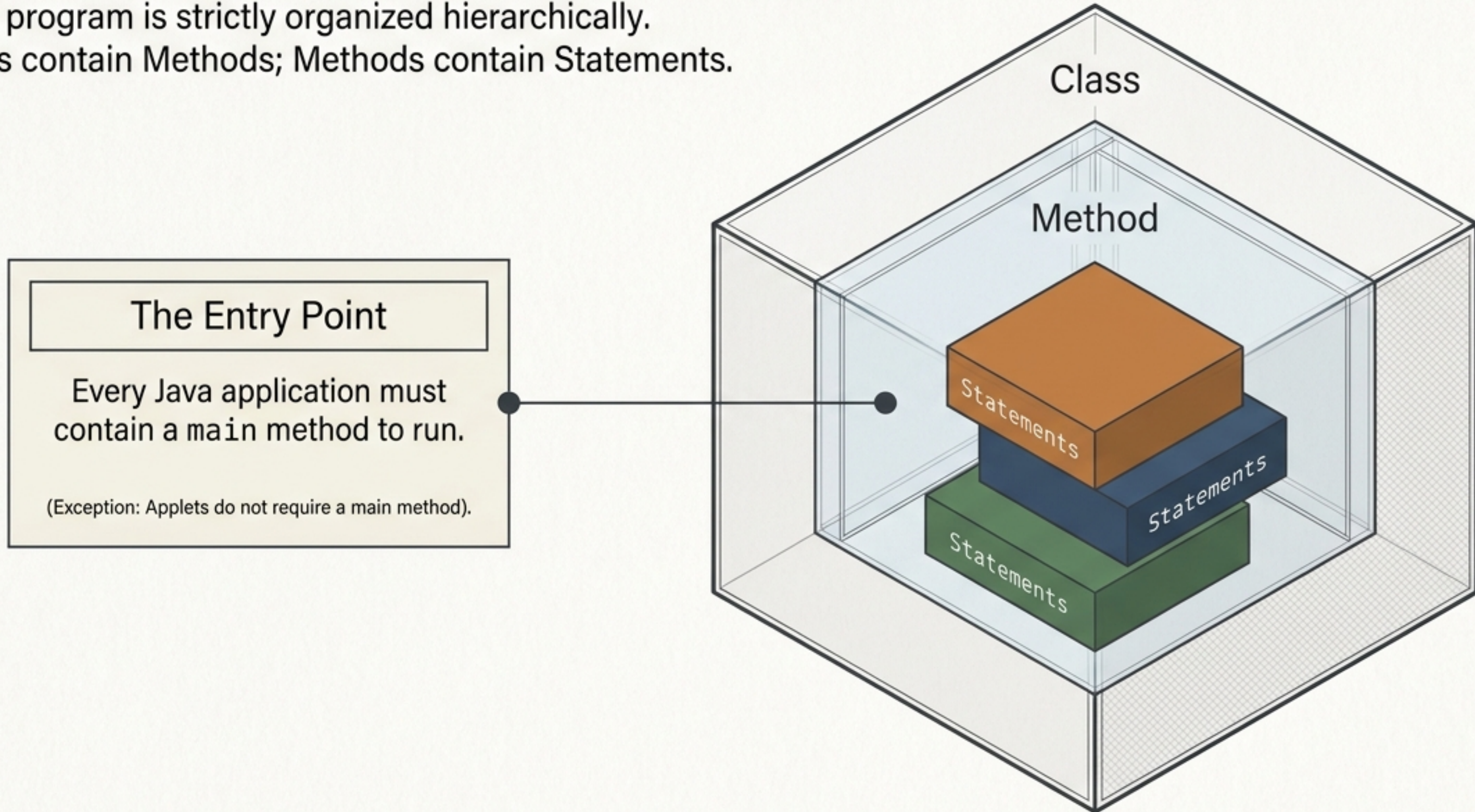


Java Fundamentals: The Architecture of Code

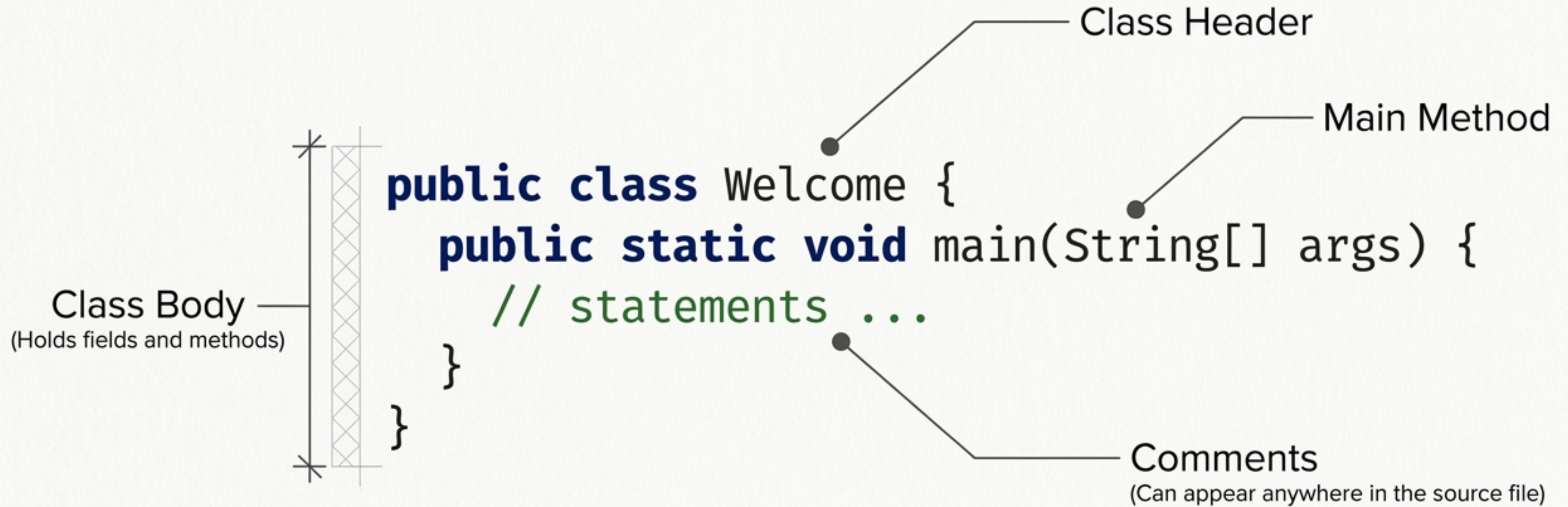
A visual breakdown of program structure,
memory allocation, and syntax.

The Macro Hierarchy: Everything Lives in a Box

A Java program is strictly organized hierarchically.
Classes contain Methods; Methods contain Statements.



Surgical Dissection Map



The Gatekeepers: Access Modifiers

Modifiers strictly control which other classes can see or use a class, method, or variable.

	Inside Same Class	Inside Same Package	Subclasses	Everywhere
public	✓	✓	✓	✓
protected	✓	✓	✓	○
default (no keyword)	✓	✓	○	○
private	✓	○	○	○

Anatomy of a Method

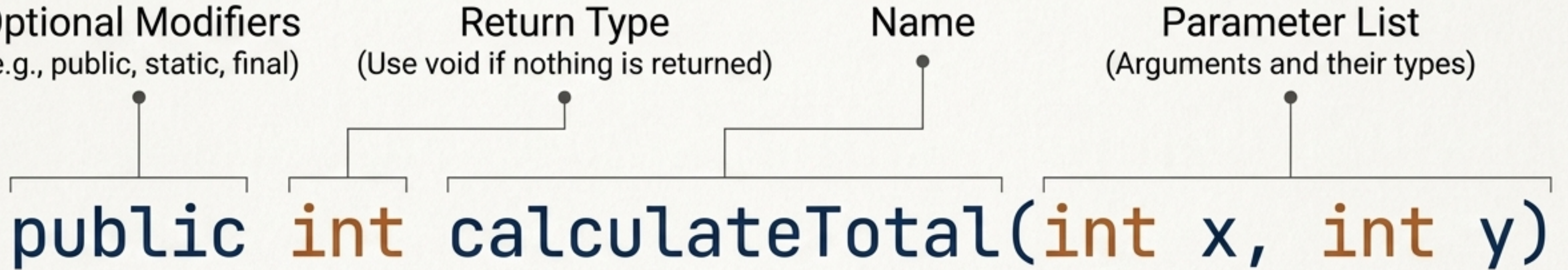
Optional Modifiers
(e.g., public, static, final)

Return Type
(Use void if nothing is returned)

Name

Parameter List
(Arguments and their types)

`public int calculateTotal(int x, int y)`



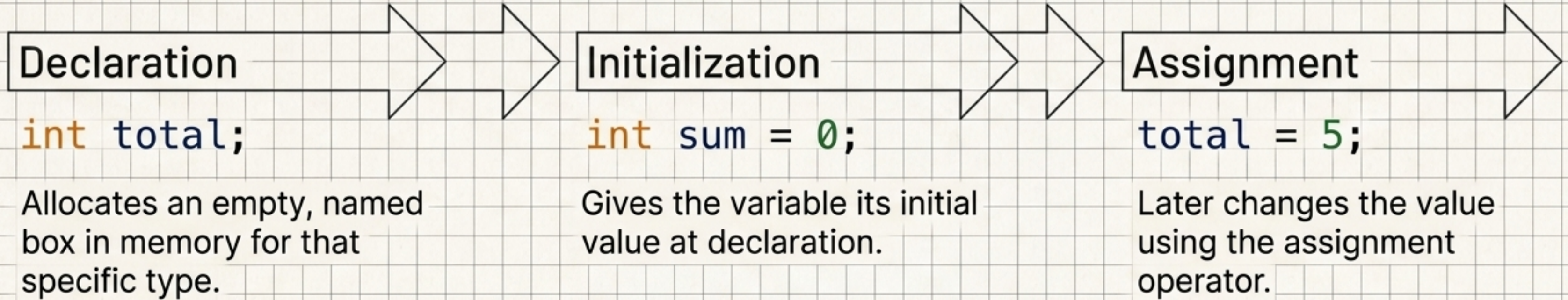
Fields

Declared inside a class.
Exist for the lifetime of the Object.

Local Variables

Declared inside a method.
Exist only while the method runs.

Variables: Allocating Memory Locations



Primitive Variable



Stores a single exact value.

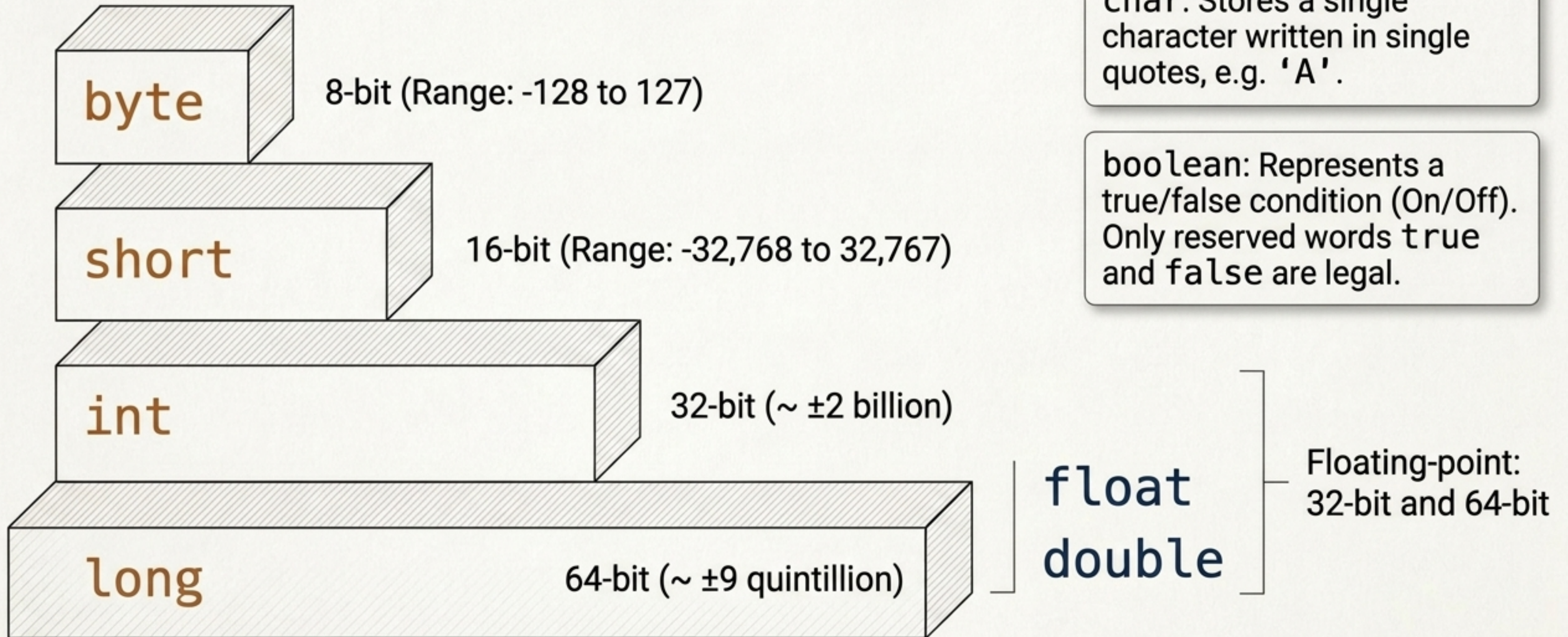
Reference Variable



Stores the memory address of an external object.

Primitive Data Types: The Memory Ladder

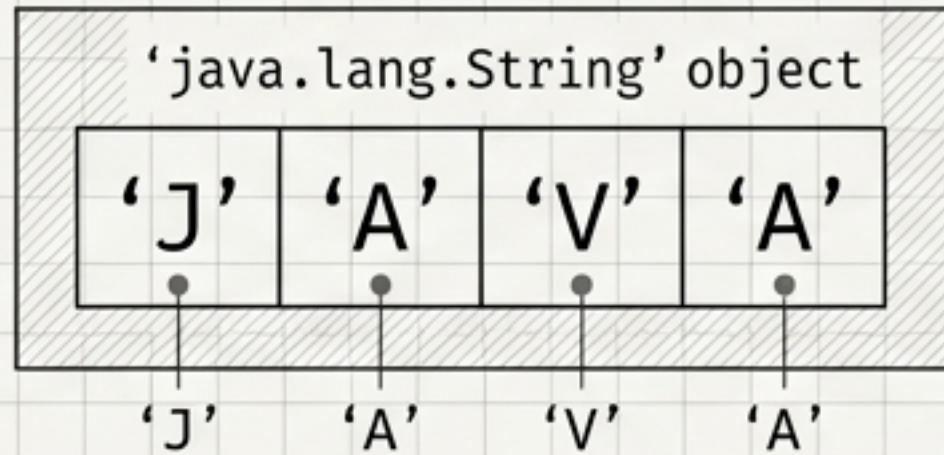
Built-in, keyword-named types that store a single exact value.



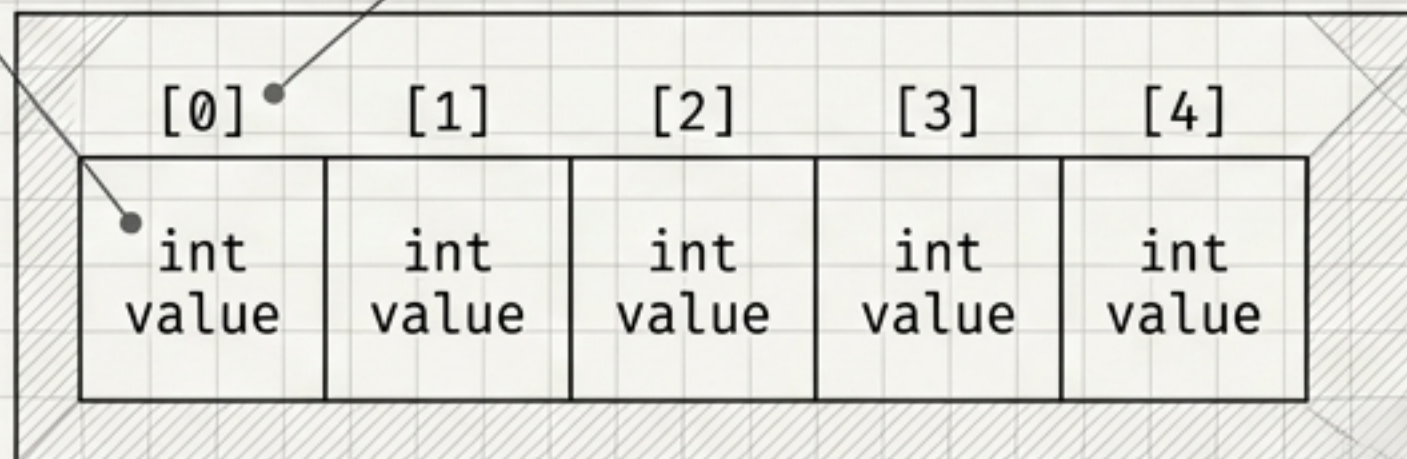
Reference (Non-Primitive) Types

Includes Classes, Objects, Interfaces, and Arrays.

1. String: A sequence of characters (an object of the class `java.lang.String`).

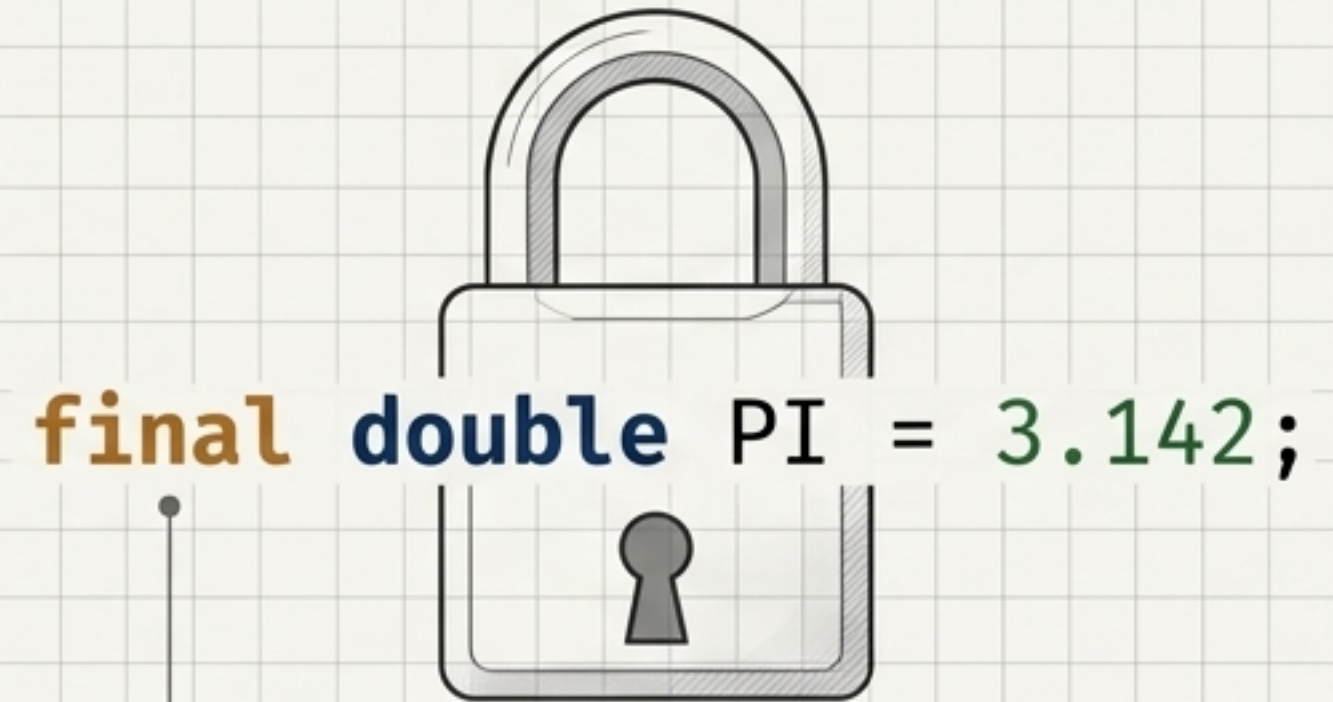


2. Array: A homogeneous collection (all elements are the exact same type), accessed by index.



Constants

The `final` reserved word ensures the value never changes during execution.



Any later assignment triggers a compile-time error. Improves readability and safer maintenance.

The Grammar of Java: Identifiers

The Rules

- ✓ Combination of letters, digits, `_`, and `$`
- ✓ Case-sensitive
- ✓ No spaces (must be a single word)
- ✓ Cannot start with a digit
- ✓ Cannot be a reserved keyword (e.g., `public`, `void`)

The Validator

<code>totalSum</code> └─ (Chosen by programmer)	✓
<code>String</code> └─ (Chosen by library author)	✓
<code>1stValue</code> └─ (Rule broken: Starts with digit)	✗
<code>class</code> └─ (Rule broken: Reserved keyword)	✗

Whitespace is Invisible

`Spaces`, `tabs`, and `blank lines` separate tokens. The Java compiler ignores extra consecutive `whitespace` entirely.

Structural Types: Classes vs. Interfaces

The Class

The Blueprint

Describes the specific behavior of an object.

Contains actual fields (variables).

Contains methods with full execution bodies.

The Interface

The Contract

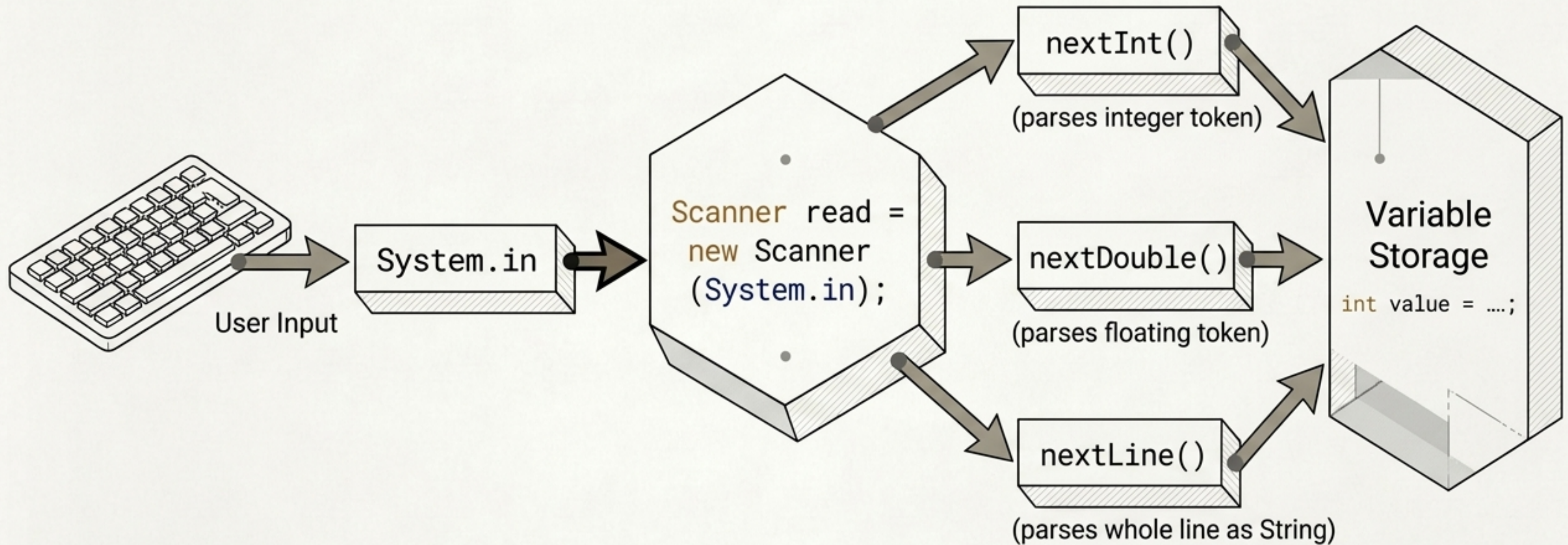
Contains method signatures and variables.

Methods are abstract by default.

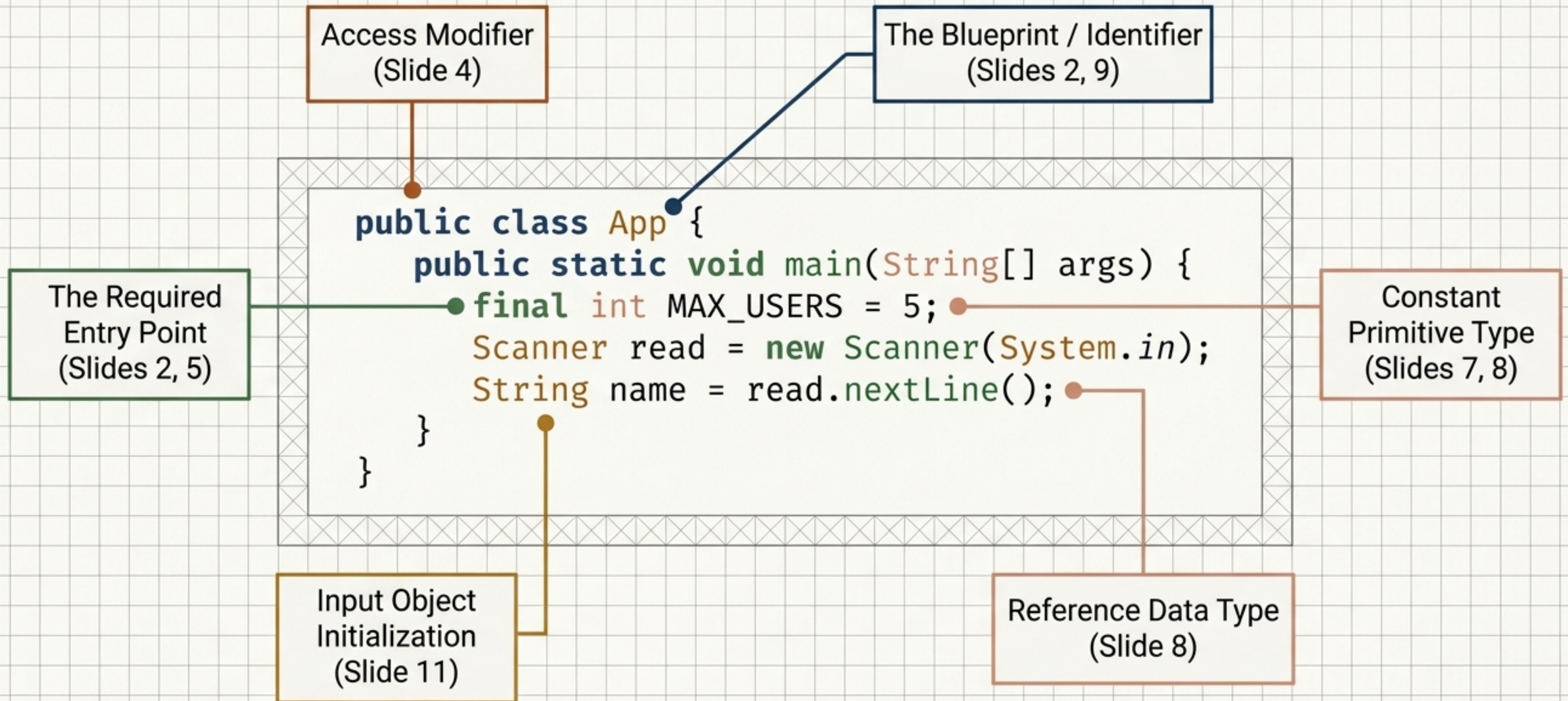
Provides the signature only, no execution body.

Interactivity: Reading User Input

The Scanner class (`java.util`) acts as the engine that processes raw input streams into usable data types.



Synthesis: The Complete Ecosystem



The Java Fundamentals Quick-Check Matrix

Architecture & Methods	Memory & Types	Syntax & Tools
Classes contain Methods; Methods contain Statements.	Variables must be declared (allocates memory).	Identifiers: case-sensitive, no spaces, no starting digit, not a keyword.
<code>main(String[] args)</code> is mandatory (except applets).	Primitive (1 value) vs Reference (Object address).	
Modifiers: <code>public</code>, <code>private</code>, <code>protected</code>, <code>default</code>.	Integer Bits: <code>byte</code> (8), <code>short</code> (16), <code>int</code> (32), <code>long</code> (64).	Scanner: <code>new Scanner(System.in)</code> → <code>nextInt()</code>, <code>nextLine()</code>.
	<code>boolean</code> (<code>true/false</code>), <code>char</code> ('single quotes').	
	<code>final</code> creates a constant.	